

■ Rainbow Grocery

Inventory Management System — Project Summary
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■ System Overview

Item	Detail
Live URL	https://rainbowv1-production-3a26.up.railway.app
Repository	github.com/SaharNoman/RainbowV1
Hosting	Railway.app — Singapore Region
Backend	Python + FastAPI + Uvicorn
Database	SQLite
Frontend	HTML + JavaScript Dashboard
Auto Deploy	Every git push triggers redeploy

■ Data & Database

Item	Detail
Database	SQLite with 2 tables — inventory and sales
Stores	11 stores: WT, DFNR, BTL, EME, GUJ, MT, SHKP, RWP, SHDR, CW, BRL
Stock Records	385 records across all stores
Sales Records	5,533 records — 13 months (MAR-25 to MAR-26), 44 articles
Auto Import	Data loads automatically from Excel on server startup

■ Dashboard — Stock Inventory Alerts

Alert	Trigger	Display
Low Stock	current_stock < threshold	Row turns red, ■ LOW badge
Store Sidebar Badge	Any low stock item in store	Red counter e.g. 4 low
Alert Banner	Any low stock in current view	Red banner at top of panel
Summary Card	Total low stock count	Red stat card

Health %	Ratio of ok vs low items	Summary card percentage
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■ Dashboard — Sales Data Alerts

Alert	Trigger	Display
Low Sales	quantity < threshold	Row turns red, ■ LOW badge
No Threshold Set	threshold = 0	Grey 'No threshold' badge
Auto Threshold	System calculated (3-month avg)	Green ■ Auto badge
Manual Override	Edited by user	Yellow — Manual badge
Reset to Auto	User clicks Reset button	Reverts to 3-month average
Store Sidebar Badge	Any low sales in store	Blue counter in sidebar
Alert Banner	Any low sales in view	Blue banner at top

■ Threshold Logic

Type	How It Works
Stock Threshold	Set manually per item per store
Sales Threshold (Auto)	Average of last 3 months: DEC-25, JAN-26, FEB-26
Manual Override	User can change any threshold via Save button
Reset to Auto	Reset button reverts threshold to 3-month average

■ Email Notifications

Manual Trigger — Send Alert Now Button

Clicking the ■ Send Alert Now button in the dashboard sends two separate emails immediately:

Email 1	Low Stock Alert	■ Low Stock Alert — X items need reorder	All items below stock threshold, grouped by store. Shows item name, price, and quantity.
Email 2	Low Sales Alert	■ Low Sales Alert — X items below sales threshold	All items below sales threshold for latest month. Shows article, quantity, and price.

Automatic Trigger — Daily at 8:00 PM

Setting	Value
Schedule	Every day at 8:00 PM automatically
Sends	Both stock + sales alert emails
Coverage	All 11 stores regardless of dashboard selection
Sender	rainbowlertsystem@gmail.com

Recipient	sahar.noman.javed@gmail.com
Protocol	Gmail SMTP SSL

■ Planned Future Features

Feature	Description
WhatsApp Notifications	Via Twilio API alongside email alerts
Sales Drop % Alert	Flag items with 30%+ drop vs previous month
Zero Sales Alert	Items with 0 sales this month across any store
Slow Moving Stock	High stock + low sales cross-reference alert
User Login & Roles	Head Office vs Store Manager access control
Excel Re-import Button	Refresh data from dashboard without code changes
Export Reports	Download low stock reports as PDF or Excel
Mobile Responsive UI	Dashboard optimized for phones and tablets